

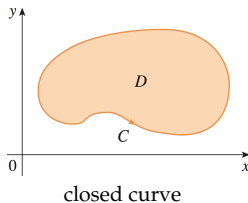
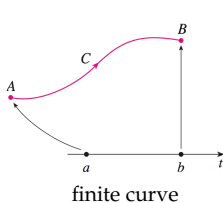
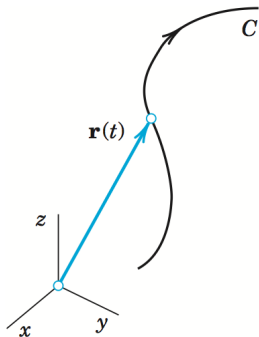
# Parametric Curves & Integration Review

## I. Parametric Curves

Defn: A **parametric curve** is a curve in space that depends on a single variable which we called the parameter. In the figure below,  $t$  is the parameter.

$$\mathbf{r}(t) = [x(t), y(t), z(t)] = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

- If  $t$  is defined on a finite interval  $[a, b]$  then  $C$  will be a finite curve.
- If  $t$  is defined on an infinite interval, say  $[0, \infty]$ , then  $C$  will be an infinite or closed curve.



Note: Defining a curve parametrically gives the curve an orientation (direction).

Find  $\mathbf{r}(t)$  by parametrizing the following curves

Ex:  $x = y^2 + y - 15, z = 0$  (Intersection of a parabolic cylinder and plane )

The easy parameterization is to let  $t = y$ :

$$\begin{aligned} x &= t^2 + t - 15 \\ y &= t \\ z &= 0 \end{aligned} \quad \Rightarrow \quad \mathbf{r}(t) = [t^2 + t - 15, t, 0] \quad t \in \mathbf{R}$$

Ex:  $4x - y = -1, 5x - z = 0$  (Intersection of 2 planes )

Let  $t = x$  and solve  $y$  &  $z$  in terms of  $x$ :

$$\begin{aligned} x &= t \\ y &= 4t + 1 \\ z &= 5t \end{aligned} \quad \Rightarrow \quad \mathbf{r}(t) = [t, 4t + 1, 5t] \quad t \in \mathbf{R}$$

Ex:  $x^2 + y^2 = 2, z = -2$  (Intersection of a cylinder and plane - circle of radius  $\sqrt{2}$   $z = -2$  )

Let  $t$  be the polar angle in the  $xy$ -plane:

$$\begin{aligned} x &= \sqrt{2} \cos t \\ y &= \sqrt{2} \sin t \\ z &= -2 \end{aligned} \quad \Rightarrow \quad \mathbf{r}(t) = [\sqrt{2} \cos t, \sqrt{2} \sin t, -2] \quad t \in [0, 2\pi]$$

Ex:  $x = y^3 + y$ ,  $z = x - y$  (Intersection of a cubic cylinder and plane )

The easy parameterization is to let  $t = y$ :

$$\begin{aligned} x &= t^3 + t \\ y &= t \\ z &= t^3 + t - t = t^3 \end{aligned} \quad \Longrightarrow \quad \mathbf{r}(t) = [t^3 + t, t, t^3] \quad t \in \mathbf{R}$$

Ex:  $x^2 + y^2 = 16$ ,  $z = 2x + y$  (Intersection of a cylinder and plane )

Let  $t$  be the polar angle in the  $xy$ -plane:

$$\begin{aligned} x &= 4 \cos t \\ y &= 4 \sin t \\ z &= 2 \cos t + \sin t \end{aligned} \quad \Longrightarrow \quad \mathbf{r}(t) = [4 \cos t, 4 \sin t, 8 \cos t + 4 \sin t] \quad t \in [0, 2\pi]$$

Ex: Find the velocity function given the position function  $\mathbf{r}(t) = [1 + t^3, te^{-t}, \sin 2t]$

Recall: *velocity*  $= \mathbf{r}'(t) = [x'(t), y'(t), z'(t)]$

$$x'(t) = 3t^2$$

$$y'(t) = e^{-t} - te^{-t}$$

$$z'(t) = 2 \cos(2t)$$

$\implies$

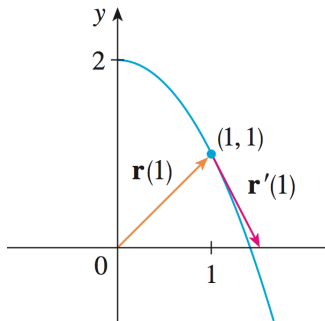
$$\mathbf{r}'(t) = [3t^2, e^{-t} - te^{-t}, 2 \cos(2t)]$$

Ex: Let  $\mathbf{r}(t) = [\sqrt{t}, 2 - t]$ . Find  $\mathbf{r}(1)$  and  $\mathbf{r}'(1)$

$$\mathbf{r}(1) = [\sqrt{1}, 2 - 1] = [1, 1]$$

$$\mathbf{r}'(t) = \left[ \frac{1}{2}t^{-1/2}, -1 \right]$$

$$\mathbf{r}'(1) = \left[ \frac{1}{2}(1)^{-1/2}, -1 \right] = \left[ \frac{1}{2}, -1 \right]$$



**Parameterizing a straight line curves.** This is simple if you know:

- (1) a vector,  $\mathbf{v}$ , parallel to the line and
- (2) a point,  $P$ , on the line.

Then the parameterization is given by:

$$\mathbf{r}(t) = \mathbf{v}t + P \quad \text{where} \quad t \in \mathbf{R}$$

Ex: Suppose we know that a line is parallel to  $\langle 1, -1, 0 \rangle$  and passes through  $(0, 5, 5)$ . Then the parameterization for this line is:

$$\mathbf{r}(t) = \langle 1, -1, 0 \rangle t + (0, 5, 5) \quad \text{or} \quad \mathbf{r}(t) = \langle t, -t + 5, 5 \rangle \quad t \in \mathbf{R}$$

Ex: Suppose we instead know that a line passes through the points  $(-3, 1, 9)$  and  $(-2, 2, 5)$ . Then we first find the vector parallel to the line by:

$$\mathbf{v} = (-3, 1, 9) - (-2, 2, 5) = \langle -1, -1, 4 \rangle$$

and so the parameterization is

$$\mathbf{r}(t) = \langle -1, -1, 4 \rangle t + (-3, 1, 9) \quad \text{or} \quad \mathbf{r}(t) = \langle -t - 3, -t + 1, 4t + 9 \rangle \quad t \in \mathbf{R}$$

**Parameterizing a straight line segments.** In this case we need just the initial and end points of the segment, say  $P$  and  $Q$ .

Then the parameterization is given by:

$$\mathbf{r}(t) = (Q - P)t + P \quad \text{where} \quad t \in [0, 1]$$

Ex: Suppose we wanted to parameterize a line with initial point  $(-1, -1, -9)$  and end point  $(10, 0, 15)$ . Then the parameterization is given by:

$$\begin{aligned}\mathbf{v} &= \langle 10, 0, 15 \rangle - \langle -1, -1, -9 \rangle + t \langle -1, -1, -9 \rangle && \text{where } t \in [0, 1] \\ &= \langle 11, 1, 24 \rangle + t \langle -1, -1, -9 \rangle \\ &= \langle 11t - 1, t - 1, 24 - 9t \rangle\end{aligned}$$

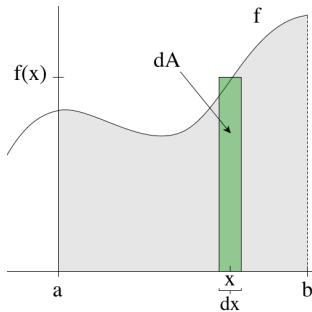
## II. Review: Integration

Remember, integration is another way of saying “add up a lot of small areas to get a whole area”. So let's think about it as such:

► Let  $dA$  be a small area (of a rectangle) and  $A$  the whole area.

► then  $dA = f(x) \cdot dx$  (see figure) and so

► 
$$A = \int_a^b dA = \int_a^b f(x) dx$$



Note: The limits of integrations ( $a$  &  $b$ ) tell you the range of the small pieces (in this example,  $dx$ )

### Interpreting Integrals: Identify the physical meaning of the Integral

- $\iiint_E \rho(x, y, z) dV$  where  $E$  is a solid sphere and  $\rho(x, y, z)$  is the density at any point in  $E$

Total Mass of  $E$

- $\int_a^b f(x) dx$  where  $f(x)$  is the force required to pull a wagon from  $a$  to  $b$

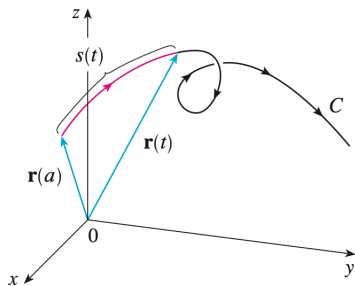
Total Work required to pull the wagon a distance  $b - a$

- $\int_C |\mathbf{v}(t)| dt$  where  $\mathbf{v}(t)$  is the velocity of a particle traveling on a finite curve  $C$

Total distance traveled by the particle along  $C$  (i.e. the arclength of  $C$ )



### III. Arclength of a Curve



Derive the integral expression for  $s(t)$ .

Assume  $C$  is parametrized by  $\mathbf{r}(t)$  for  $a \leq t \leq b$

We want to think of adding up many “small” line segments, so that we can use an integral

$$s = \int_C ds$$

But, what is  $ds$ ? ( $ds$  is a “small” distance)

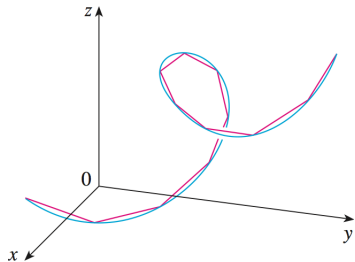
Recall: distance = (rate)(time), so

“small” distance = (rate) (“small” time)

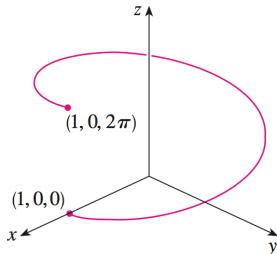
$$\text{or } ds = |\mathbf{r}'(t)| dt$$

Therefore

$$s(t) = \int_a^b |\mathbf{r}'(t)| dt$$



Compute the arclength of the curve:  $\mathbf{r}(t) = [\cos t, \sin t, t]$



Need:  $\mathbf{r}'(t)$  and  $[a, b]$  (Range of the parameter  $t$ )

Note: Since  $z = t$ , then the range of  $t$ -values is the same as the range of  $z$ -values

$$\Rightarrow t \in [0, 2\pi] \quad (\text{This is } [a, b])$$

$$\mathbf{r}'(t) = [-\sin t, \cos t, 1]$$

$$\|\mathbf{r}'(t)\|^2 = \sin^2 t + \cos^2 t + 1^2 = 1 + 1 = 2 \quad \Rightarrow \quad \|\mathbf{r}'(t)\| = \sqrt{2}$$

$$s(t) = \int_a^b |\mathbf{r}'(t)| dt = \int_0^{2\pi} \sqrt{2} dt = \sqrt{2} t \Big|_0^{2\pi} = 2\sqrt{2}\pi$$